Asymptotic Methods

Part II

Recorded from the lecturer in 2026-2027.

Pre-requisites

- Complex methods / analysis.
- Refer to Section II in Stuart.

1. Intro & Motivation

- Most real problems don't have closed form solutions, or if they do, they are unworkable or difficult to interpret.
- Numerical approximations can be useful, but they can be expensive, they may have bad numerical properties, or solutions may be inscrutable.
- Taylor series, etc. can be limited in applicability.
- Asymptotic methods emphasise usefulness, interpretability as well as accuracy.

Examples

e.g. 1)

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t}\, dt, \qquad \mathrm{Re}(z) > 0.$$

For n an integer,

$$\Gamma(n+1) = n! \sim \sqrt{2\pi}\, n^{n+\frac{1}{2}} e^{-n} \left(1 + \frac{1}{12n} + \frac{1}{288n^2} + \cdots\right) \qquad \text{as } n \to \infty.$$

e.g. 2)

$$x^2 y'' + xy' - x^2 y = 0.$$

Exact series solution:

$$I_0(x) = \sum_{n=0}^{\infty} \frac{x^{2n}}{2^{2n}(n!)^2}.$$

Liouville–Green:

$$\implies I_0(x) \sim (2\pi x)^{-1/2} e^x \qquad \text{as } x \to \infty.$$

e.g. 3)

$$y'' + k^2 e^{-x^2/2} y = 0,$$

with

$$y \to y_0 \qquad \text{as } x \to \infty,$$

$$y \to \pm y_0 \qquad \text{as } x \to -\infty.$$

$$k = ?, \qquad k = k_0, k_1, k_2, \ldots$$

$$\implies k_n \sim \frac{\sqrt{\pi}}{2}\left(n + \frac{1}{2}\right) \qquad \text{as } n \to \infty.$$

WKBJ.

- Convergent methods typically give arbitrarily accurate approximations with more & more terms/iterations.
- Asymptotic methods give arbitrarily accurate approximations as some variable approaches a limit. Thus they always come with a qualifier, e.g. $z \to 0$ or $n \to \infty$.

2. Asymptotic Expansions

2.1 Definitions & Notation

(a) Asymptotic Equality/Equivalence

For 2 functions $f(x)$ and $g(x)$, we have

$$f(x) \sim g(x) \qquad \text{about } x = x_0$$

if

$$\boxed{\lim_{x \to x_0} \frac{f(x)}{g(x)} = 1.}$$

Remarks:

- Usually we want to approximate f with g.
- We can write
$$f(x) = \underbrace{g(x)}_{\text{dominant}} + \underbrace{[f(x) - g(x)]}_{\text{subdominant}}.$$
- Note $\sim$ is an equivalence relation (that is reflexive, transitive, symmetric).
- It respects products:
$$f_1 \sim g_1, \qquad f_2 \sim g_2$$
$$\implies f_1 f_2 \sim g_1 g_2.$$
- It is linear:
$$\alpha f_1 + \beta f_2 \sim \alpha g_1 + \beta g_2,$$
where α, β are constants.
- Definition has problems when dealing with oscillatory functions that go through 0 as $x \to x_0$.

(b) "Little o"

Suppose f & g are real functions.

We write

$$\boxed{f(x) = o(g(x)) \qquad \text{as } x \to x_0}$$

if

$$\boxed{\lim_{x \to x_0} \frac{f(x)}{g(x)} = 0.}$$

Remark:

$$f(x) = o(g(x))$$

can also be written as

$$f(x) \ll g(x) \qquad \text{as } x \to x_0.$$

- Can be framed in inequalities, e.g. $x_0 = \infty$ and let $g > 0$.
  Then $f = o(g)$ as $x \to \infty$ is equivalent to

$$\forall \varepsilon > 0, \quad \exists a \quad \text{s.t. when } x > a,$$

$$\boxed{|f| \leq \varepsilon g.}$$

(c) "Big O"

We write

$$f(x) = O(g(x)) \qquad \text{as } x \to x_0$$

for $g > 0$, if

$$\exists C > 0 \quad \text{s.t.}$$

$$\boxed{|f(x)| \leq C g(x)}$$

for x sufficiently close to x_0.

Warning:

In some areas of physics people use big-O and $\sim$ interchangeably.

Little Exercises: Show That

(a)

$$f(x) \sim g(x) \iff f - g = o(g) \qquad \text{as } x \to x_0.$$

(b)

$$\sin x \ll x^2 \qquad \text{as } x \to \infty.$$

(c)

$$12x^5 + 6x^3 + x = O(x^5) \qquad \text{as } x \to \infty.$$

(d)

$$e^{ax} \ll e^{bx} \qquad \text{as } x \to \infty,$$

if $b > a > 0$.

(e)

$$e^x \gg x^N \qquad \text{as } x \to \infty,$$

$$\forall N \in \mathbb{Z}^+.$$

2.2 Asymptotic Expansions

- Have unworkable/unknown $f(x)$.
- Want to know its behaviour in a limit.
- Want to find a simpler/known $g(x)$ so that

$$f \sim g.$$

- Also want to break up g into a hierarchy of approximations, s.t.

$$g(x) = \sum_{k=0}^{N} a_k \phi_k(x),$$

where a_k are coefficients, ϕ_k are known & workable, and where adding more terms gives a better approximation that can be estimated.

(a) Asymptotic Sequence

A sequence of functions

$$\{\phi_0(x), \phi_1(x), \ldots\}$$

is an "asymptotic sequence" as $x \to x_0$ if

$$\boxed{\phi_{k+1}(x) \ll \phi_k(x) \qquad \text{as } x \to x_0, \quad \forall k.}$$

e.g. 1)

$$\phi_k = (x-a)^k \qquad \text{around } x = a.$$

Check:

$$\frac{\phi_{k+1}}{\phi_k} = x - a \to 0 \qquad \text{as } x \to a. \quad \checkmark$$

e.g. 2)

$$\phi_k = e^{-kx} x^k \qquad \text{as } x \to \infty \text{ or } x \to -\infty.$$

Check:

$$\frac{\phi_{k+1}}{\phi_k} = e^{-x} x.$$

$$e^{-x} x \to 0 \quad \text{as } x \to \infty \qquad \checkmark$$
$$e^{-x} x \to \infty \quad \text{as } x \to -\infty \qquad \times$$

e.g. 3)

$$\phi_k = \cos(nx) \qquad \text{as } x \to \infty.$$

$\times$

e.g. 4)

$$\{|x|, \log|x|, \log|\log|x||, \ldots\} \qquad \text{as } x \to \infty.$$

?

(b) Asymptotic Expansion

- Suppose $f(x)$ is a function we want to understand near $x = x_0$.
- Let $\{\phi_k\}_{k=0}^{\infty}$ be an asymptotic sequence.

Then, the finite sum

$$\sum_{k=0}^{N} a_k \phi_k(x)$$

for some coefficients, is an (N-term) asymptotic approximation of $f(x)$ as $x \to x_0$ if:

$$\boxed{f(x) = \sum_{k=0}^{N} a_k \phi_k(x) + o(\phi_N(x))} \qquad \text{as } x \to x_0.$$

Alternatively, we can write the above as

$$\boxed{\frac{f(x) - \sum_{k=0}^{N} a_k \phi_k(x)}{\phi_N(x)} \longrightarrow 0} \qquad \text{as } x \to x_0.$$

End of lecture 1

(c) Asymptotic expansions/series

If (b) holds for *all* N, then this is called an "asymptotic series or expansion", and we write

$$f(x) \sim \sum_{k=0}^{\infty} a_k \phi_k(x) \qquad \text{as } x \to x_0.$$

Attention: No claim of convergence is made. It is just notation. We never fix x and then take $N \to \infty$.

Instead, we have an infinite number of approximations (of different N), and $x \to x_0$ is taken in each separately!

(d) Properties of asymptotic expansions and remarks

(I) Mathematical operations

Suppose $\{\phi_k(x)\}$ is an asymptotic sequence as $x \to x_0$, and let

$$f \sim \sum_{k=0}^{\infty} a_k \phi_k, \qquad g \sim \sum_{k=0}^{\infty} b_k \phi_k.$$

Linearity

$$\alpha f + \beta g \sim \sum_{k=0}^{\infty} (\alpha a_k + \beta b_k) \phi_k \qquad \text{as } x \to x_0.$$

Multiplication

If $\phi_k = (x - x_0)^k$, then

$$fg \sim \sum_{k=0}^{\infty} c_k \phi_k, \qquad c_k = \sum_{j=0}^{k} a_j b_{k-j}.$$

Division

See 3.8 BO.

Integration

Term-by-term integration is usually okay.

For example, if $\phi_k = (x - x_0)^k$, then

$$\int_{x_0}^{x} f(t)\,dt \sim \sum_{k=0}^{\infty} \frac{a_k (x - x_0)^{k+1}}{k+1} \qquad \text{as } x \to x_0.$$

Exercise: Use the definitions to prove this.

Differentiation

Term-by-term differentiation is not always okay.

To do this, you need $f'(x)$ to be continuous and to have an asymptotic series, i.e.

$$f'(x) \sim \sum_{k=0}^{\infty} c_k \phi_k(x).$$

(II) Different asymptotic representations

Suppose $\{\phi_k\}$ and $\{\psi_k\}$ are different asymptotic sequences as $x \to x_0$. Also,

$$f \sim \sum_{k=0}^{\infty} a_k \phi_k, \qquad f \sim \sum_{k=0}^{\infty} b_k \psi_k \qquad \text{as } x \to x_0.$$

Then a_k and b_k are not the same.

For example,

$$f(x) = 1 + \frac{1}{x}.$$

Consider $\phi_k = x^{-k}$ as $x \to \infty$. Then

$$f(x) \sim \sum_{k=0}^{\infty} a_k \phi_k, \qquad a_0 = 1, \quad a_1 = 1, \quad a_n = 0 \quad (n > 1).$$

Now consider

$$\psi_k = (1 + x)^{-k}.$$

Then

$$\begin{aligned} f = 1 + \frac{1}{x} &= \frac{1}{1 - (x+1)^{-1}} \\ &= 1 + \frac{1}{1+x} + \frac{1}{(1+x)^2} + \cdots \\ &\sim \sum_{k=0}^{\infty} b_k \psi_k, \end{aligned}$$

where

$$b_k = 1 \qquad \forall k.$$

(III) Uniqueness of coefficients

Let $\{\phi_k\}$ be an asymptotic sequence, and

$$f(x) \sim \sum_{k=0}^{\infty} a_k \phi_k \qquad \text{as } x \to x_0.$$

Then the coefficients a_k are unique.

In fact, the a_k can be constructed:

$$a_0 = \lim_{x \to x_0} \frac{f(x)}{\phi_0(x)},$$

$$a_n = \lim_{x \to x_0} \left[\frac{f(x) - \sum_{k=0}^{n-1} a_k \phi_k(x)}{\phi_n(x)} \right].$$

Proof. First, we know

$$\lim_{x \to x_0} \frac{f(x) - \sum_{k=0}^{N} a_k \phi_k}{\phi_N} = 0 \qquad \forall N.$$

Let $N = 0$. Then

$$\lim_{x \to x_0} \frac{f(x) - a_0 \phi_0}{\phi_0} = 0,$$

$$\lim_{x \to x_0} \left(\frac{f(x)}{\phi_0} \right) - a_0 = 0.$$

Use induction on N.

Suppose the formulae are true for $N = n$, and the coefficients are unique for $k \leq n$. Then

$$\lim_{x \to x_0} \frac{f(x) - \sum_{k=0}^{n} a_k \phi_k}{\phi_n} = 0$$

$$= \lim_{x \to x_0} \left(\frac{f(x) - \sum_{k=0}^{n-1} a_k \phi_k}{\phi_n} \right) - a_n.$$

Alternative proof: see p. 3 in Manton.

(IV) An asymptotic series does not uniquely determine the function it is describing

For example, suppose

$$f(x) \sim g(x) \qquad \text{as } x \to x_0,$$

and

$$h(x) \ll f(x) \qquad \text{as } x \to x_0.$$

Then

$$f(x) + h(x) \sim g(x) \qquad \text{as } x \to x_0.$$

So, if g is an asymptotic expansion, it can describe both f and $f+h$.

For example,

$$f_1 = \frac{1}{1+x}, \qquad f_2 = \frac{1+e^{-x}}{1+x}.$$

Both are

$$\sim \sum_{k=0}^{\infty} (-1)^k x^{-(k+1)} \qquad \text{as } x \to \infty.$$

(V) Power series

In this course, we mostly take

$$\phi_k = (x - x_0)^k \qquad \text{as } x \to x_0,$$

or

$$\phi_k = x^{-k} \qquad \text{as } x \to \infty.$$

In fact, a Taylor series about a point is an asymptotic series about that point:

$$f(x) - \sum_{k=0}^{N} \frac{f^{(k)}(a)(x-a)^k}{k!} = o\big(|x-a|^N\big)\,.$$

Exercise: Prove this using the definitions and the integral form of the remainder term.

Terminology: A term is said to be "small beyond all orders" in some limit if it cannot be represented by any power series or, equivalently, is smaller than any power.

For example, e^{-x} is small beyond all orders in inverse powers of x as $x \to \infty$:

$$e^{-x} \ll x^{-k} \qquad \forall k, \qquad \text{as } x \to \infty.$$

(VI) Optimal truncation

Question: If an asymptotic series diverges, how many terms should I include in the sum?

If

$$f(x) \sim \sum_{n=0}^{\infty} a_n \psi_n(x) \qquad \text{as } x \to x_0,$$

and we want to get the best approximation to $f(x)$ at $x = x_1$, where x_1 is close to x_0, then truncate the sum at the smallest term (or the one before).

So stop adding terms when

$$|a_k \psi_k(x_1)| < |a_{k+1} \psi_{k+1}(x_1)|.$$

This gives you an unexpectedly excellent approximation.

End of lecture 2

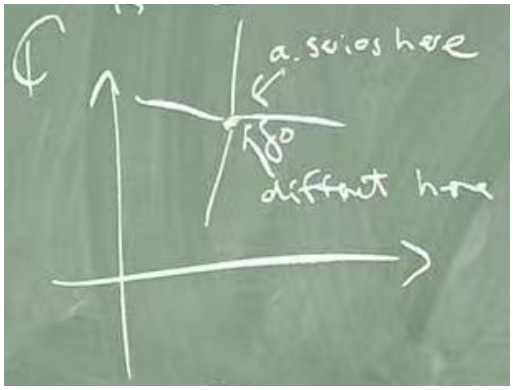
2.3 A. expansions in $\mathbb{C}$ — Stokes phenomenon

- So far our defns relied on real limits, $x, x_0 \in \mathbb{R}$ etc.
- When we use complex variables, $x = z \in \mathbb{C}$ and $z \to z_0$, typically z must be restricted to a sector of $\mathbb{C}$ around z_0 for some of our definitions to work.

  [See III.2 in Stuart]

  e.g. little o & O defns need to bring in moduli of functions.

- Even if $f(z)$ analytic at z_0, it may have different a. series as z_0 is approached from different sectors in $\mathbb{C}$. This is Stokes phenomenon.



e.g.

$$f(z) = \sinh(z^2) = \frac{1}{2}\left(e^{z^2} - e^{-z^2}\right), \qquad z \in \mathbb{C}.$$

What is its behaviour as $|z| \to \infty$?

Which of e^{z^2} or e^{-z^2} is dominant?

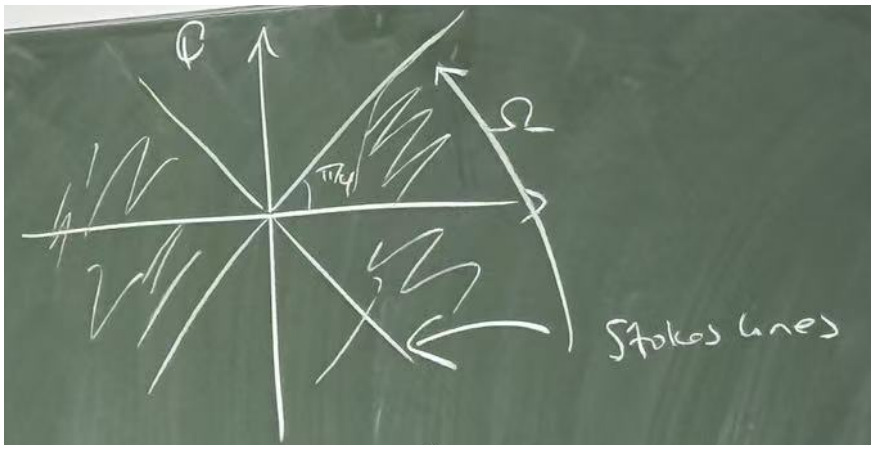
$$\left|\frac{e^{-z^2}}{e^{z^2}}\right| = \left|e^{-2z^2}\right| = e^{-2\,\mathrm{Re}(z^2)}.$$

(a) Suppose $|z| \to \infty$ & $\mathrm{Re}(z^2) > 0$.

$$\text{ratio} \to 0 \quad \Rightarrow \quad e^{-z^2} \ll e^{z^2} \quad \Rightarrow \quad f(z) \sim \frac{1}{2}e^{z^2} \quad \text{in this limit.}$$

$$\mathrm{Re}(z^2) > 0 \quad \Rightarrow \quad \Omega: \quad -\frac{\pi}{4} + \delta < \arg z < \frac{\pi}{4} - \delta, \quad \text{and} \quad \frac{3\pi}{4} + \delta < \arg z < \frac{5\pi}{4} - \delta,$$

for small δ.



(b) $|z| \to \infty$ & $\mathrm{Re}(z^2) < 0$.

$$\text{ratio} \to \infty \quad \Rightarrow \quad e^{z^2} \ll e^{-z^2} \quad \& \quad f \sim -\frac{1}{2}e^{-z^2}.$$

$$\widetilde{\Omega}: \quad \frac{\pi}{4}+\delta < \arg z < \frac{3\pi}{4}-\delta \quad \& \quad -\frac{3\pi}{4}+\delta < \arg z < -\frac{\pi}{4}-\delta.$$

$$f(z) \sim \begin{cases} \frac{1}{2}e^{z^2}, & |z|\to\infty \ \& \ z\in\Omega, \\ -\frac{1}{2}e^{-z^2}, & |z|\to\infty \ \& \ z\in\widetilde{\Omega}. \end{cases}$$

- A. behaviour of f as $|z|\to\infty$ depends on which wedge $z\to\infty$ within.
- $\mathbb{C}$ is divided into sectors, separated by special lines, called Stokes lines. [Warning, sometimes called anti-Stokes lines!!]

3. Approximations of Integrals

3.1 Taylor expanding the integrand — Stieltjes integrals

(Warm up problem)

Consider:

$$I(x) = \int_0^\infty \frac{e^{-t}}{1+xt}\,dt, \qquad x\geq 0.$$

- We want to understand I's behaviour when $x\to 0^+$.
- How about we expand $\dfrac{1}{1+xt}$?

$$\frac{1}{1+xt} = \sum_{n=0}^{\infty}(-1)^n x^n t^n \qquad \times \quad t \text{ can be big!}$$

Consider instead

$$\sum_{n=0}^{N}(-xt)^n = \frac{1-(-xt)^{N+1}}{1+xt} = \frac{1}{1+xt} - \underbrace{\frac{(-xt)^{N+1}}{1+xt}}_{r_N}.$$

$$\Rightarrow \quad \frac{1}{1+xt} = \sum_{n=0}^{N}(-xt)^n + r_N.$$

$$\begin{aligned} I(x) &= \int_0^\infty e^{-t}\left[\sum_{n=0}^{N}(-1)^n x^n t^n + r_N\right]dt \\ &= \sum_{n=0}^{N}(-1)^n x^n \underbrace{\int_0^\infty t^n e^{-t}\,dt}_{=n!} + R_N \\ &= \sum_{n=0}^{N}(-1)^n n!\, x^n + R_N \\ &= \sum_{n=0}^{N} c_n \phi_n(x) + R_N. \end{aligned}$$

$\phi_n = x^n$ as $x\to 0^+$, so is a. sequence!

Need to show $R_N = o(x^N)$ as $x\to 0^+$.

$$\begin{aligned} \left|\frac{R_N}{x^N}\right| &= \frac{x^{N+1}}{x^N}\int_0^\infty \frac{t^{N+1}e^{-t}}{1+xt}\,dt \\ &\leq x\int_0^\infty t^{N+1}e^{-t}\,dt = x(N+1)! \longrightarrow 0 \quad \text{as } x\to 0^+. \end{aligned}$$

Note $1 + xt \geq 1$ for $x, t \geq 0$. True for every fixed $N \geq 0$.

$$\Rightarrow \quad I(x) \sim \sum_{n=0}^{\infty} (-1)^n n!\, x^n \qquad \text{as } x \to 0^+.$$

3.2 Integration by parts

Recap:

$$\int_a^b f'(t)g(t)\,dt = [fg]_a^b - \int_a^b fg'\,dt.$$

Generalisation:

$$\begin{aligned} I &= \int_a^b f(t)g^{(n)}(t)\,dt \\ &= \sum_{k=1}^{n} (-1)^{k-1} \left[f^{(k-1)}(t)g^{(n-k)}(t) \right]_a^b + (-1)^n \int_a^b f^{(n)}(t)g(t)\,dt. \end{aligned}$$

- Repeated integration by parts can generate an a. approximation under certain circumstances.

e.g. How does

$$F(x) = \int_x^\infty e^{-t^2/2}\,dt$$

behave as $x \to \infty$?

$$\begin{aligned} F(x) &= \int_x^\infty \frac{te^{-t^2/2}}{t}\,dt \\ &= \left[-\frac{1}{t} e^{-t^2/2} \right]_x^\infty - \int_x^\infty \frac{e^{-t^2/2}}{t^2}\,dt \\ &= \underbrace{\frac{e^{-x^2/2}}{x}}_{\text{a. approximation}} + \underbrace{\left(- \int_x^\infty \frac{e^{-t^2/2}}{t^2}\,dt \right)}_{\text{error}}. \end{aligned}$$

Need to check that

$$\text{error} = o\left(\frac{e^{-x^2/2}}{x} \right) \qquad \text{as } x \to \infty.$$

$$\begin{aligned} |\text{error}| &= \int_x^\infty \frac{te^{-t^2/2}}{t^3}\,dt \qquad (\text{note } t \geq x) \\ &\leq \frac{1}{x^3} \int_x^\infty te^{-t^2/2}\,dt = \frac{1}{x^3} e^{-x^2/2} \\ &= o\left(\frac{e^{-x^2/2}}{x} \right) \qquad \text{as } x \to \infty. \end{aligned}$$

$$\Rightarrow \quad F(x) \sim \frac{e^{-x^2/2}}{x} \qquad \text{as } x \to \infty.$$

In fact we find

$$F(x) \sim \frac{e^{-x^2/2}}{x} \sum_{n=0}^{\infty} (-1)^n \frac{(2n-1)!!}{x^{2n}} \qquad \text{as } x \to \infty.$$

$$(2n-1)!! = (2n-1)(2n-3)(2n-5)\cdots 3 \cdot 1 \quad (n \geq 1), \qquad (-1)!! = 1.$$

Exercise: Show this.

End of lecture 3

Example 2. Find the behaviour of

$$I(x) = \int_a^b f(t)e^{xt}\,dt \qquad \text{as } x \to \infty,$$

where $b > a$, $f(b) \neq 0$, and f is smooth on $[a, b]$.

As $x \to \infty$, the integral is dominated by a narrow region near $t = b$ because of the exponential. Very roughly,

$$I(x) \approx f(b) \int_a^b e^{xt}\,dt = \frac{f(b)}{x}(e^{xb} - e^{xa}) \sim \frac{f(b)e^{xb}}{x},$$

using $e^{xb} \gg e^{xa}$ as $x \to \infty$.

Repeated integration by parts, using $e^{xt} = \dfrac{d}{dt}\left(\dfrac{e^{xt}}{x}\right)$, gives

$$\begin{aligned} I(x) &= \sum_{k=1}^{N}(-1)^{k-1}\left[f^{(k-1)}(t)e^{xt}x^{-k}\right]_a^b + \frac{(-1)^N}{x^N}\int_a^b f^{(N)}(t)e^{xt}\,dt \\ &= e^{xb}\big(J_1 - J_2 + J_3\big), \end{aligned}$$

where

$$\begin{aligned} J_1 &= \sum_{k=1}^{N}(-1)^{k-1}f^{(k-1)}(b)x^{-k}, \\ J_2 &= e^{-x(b-a)}\sum_{k=1}^{N}(-1)^{k-1}f^{(k-1)}(a)x^{-k}, \\ J_3 &= \frac{(-1)^N}{x^N}\int_a^b f^{(N)}(t)e^{x(t-b)}\,dt. \end{aligned}$$

J_1 could be an N-term a. approximation with $\phi_k = x^{-k}$ as $x \to \infty$, but we need to show that

$$J_2 = o(x^{-N}), \qquad J_3 = o(x^{-N}) \qquad \text{as } x \to \infty.$$

For the first term,

$$\frac{J_2}{x^{-N}} = e^{-x(b-a)}\sum_{k=1}^{N}(-1)^{k-1}f^{(k-1)}(a)x^{N-k} \longrightarrow 0 \qquad \text{as } x \to \infty,$$

which is true for every N. For the remainder,

$$\begin{aligned} \left|\frac{J_3}{x^{-N}}\right| &\le \int_a^b |f^{(N)}(t)|e^{x(t-b)}\,dt \\ &\le \max_{a\le t\le b}|f^{(N)}(t)|\int_a^b e^{x(t-b)}\,dt \\ &= \max_{a\le t\le b}|f^{(N)}(t)|\frac{1-e^{-x(b-a)}}{x} \longrightarrow 0 \qquad \text{as } x \to \infty. \end{aligned}$$

This holds for every N for which f has N continuous derivatives. If f is infinitely differentiable, then

$$I(x) = \int_a^b f(t)e^{xt}\,dt \sim \sum_{k=1}^{\infty}(-1)^{k-1}\frac{f^{(k-1)}(b)e^{xb}}{x^k} \qquad \text{as } x \to \infty.$$

3.3 The Gamma function

Recall

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t}\, dt, \qquad \mathrm{Re}(z) > 0.$$

(a) Γ is analytic for $\mathrm{Re}(z) > 0$.

(b) $\Gamma(z+1) = z\Gamma(z)$ (recursion relation).

(c) $\Gamma(1) = 1$ and $\Gamma(n+1) = n!$ for positive integers n.

(d) For $x > 0$ and $\mathrm{Re}(z) > 0$,

$$\int_0^\infty t^{z-1} e^{-xt}\, dt = \frac{\Gamma(z)}{x^z}.$$

(e) Generalisation of binomial coefficients:

$$\binom{z}{k} = \frac{1}{k!}\frac{\Gamma(z+1)}{\Gamma(z-k+1)}, \qquad z \in \mathbb{C}, \quad k = 0, 1, 2, 3, \ldots,$$

with the quotient interpreted by continuation where necessary.

$$(1+t)^z = \sum_{k=0}^{\infty} \binom{z}{k} t^k, \qquad |t| < 1.$$

(f) Reflection and duplication formulae: see Section 3.2 in Chapling.

3.4 Watson's lemma

Consider

$$F(x) = \int_0^T f(t) e^{-xt}\, dt \qquad \text{as } x \to \infty, \qquad 0 < T \leq \infty.$$

Suppose

$$f(t) \sim t^\alpha \sum_{n=0}^{\infty} a_n t^{\beta n} \qquad \text{as } t \to 0^+, \qquad \alpha > -1, \quad \beta > 0.$$

Also assume that either

(i) $|f(t)| < Ke^{bt}$ for all $t \in [0, T]$, for some $K, b > 0$; or

(ii) $\displaystyle\int_0^T |f(t)|\, dt < \infty$.

Then

$$\boxed{F(x) \sim \sum_{n=0}^{\infty} a_n \frac{\Gamma(\alpha + \beta n + 1)}{x^{\alpha+\beta n+1}}} \qquad \text{as } x \to \infty.$$

This is *Watson's lemma*.

Example 2.

$$xy'' + (a + 1 - x)y' - ay = 0, \qquad a > 0.$$

One solution is

$$M(x) = \frac{e^x}{\Gamma(a)} \int_0^1 e^{-xt}(1-t)^{a-1}\, dt.$$

What is its behaviour as $x \to \infty$?

Here $f(t) = (1-t)^{a-1}$ is integrable, so case (ii) applies, and

$$f(t) = (1-t)^{a-1} = \sum_{n=0}^{\infty} (-1)^n \frac{\Gamma(a)}{\Gamma(a-n)n!}\, t^n, \qquad |t| < 1.$$

Thus $\alpha = 0$, $\beta = 1$, and Watson's lemma gives

$$\begin{aligned} M(x) &\sim \frac{e^x}{\Gamma(a)} \sum_{n=0}^{\infty} \frac{(-1)^n \Gamma(a)\Gamma(n+1)}{\Gamma(a-n)n!} \frac{1}{x^{n+1}} \\ &= e^x \sum_{n=0}^{\infty} \frac{(-1)^n}{\Gamma(a-n)x^{n+1}} \qquad \text{as } x \to \infty. \end{aligned}$$

In particular,

$$M(x) \sim \frac{e^x}{x\Gamma(a)} \qquad \text{as } x \to \infty.$$

Proof of Watson's lemma. Write

$$\begin{aligned} F(x) &= \int_0^{\epsilon} f(t) e^{-xt}\, dt + \int_{\epsilon}^{T} f(t) e^{-xt}\, dt \\ &\equiv F_1(x) + F_2(x), \end{aligned}$$

where $\epsilon > 0$ is small and fixed, with $\epsilon < T$.

First, show that F_2 is small.

Case (i): for $x > b$,

$$\begin{aligned} |F_2(x)| &\le \int_{\epsilon}^{T} K e^{bt} e^{-xt}\, dt \\ &\le \int_{\epsilon}^{\infty} K e^{-(x-b)t}\, dt \\ &= \frac{K}{x-b} e^{(b-x)\epsilon} \\ &= O\left(\frac{e^{-\epsilon x}}{x}\right) \qquad \text{as } x \to \infty. \end{aligned}$$

Case (ii):

$$\begin{aligned} |F_2(x)| &\le \int_{\epsilon}^{T} |f(t)| e^{-xt}\, dt \\ &\le e^{-\epsilon x} \int_{\epsilon}^{T} |f(t)|\, dt \\ &\le e^{-\epsilon x} \int_{0}^{T} |f(t)|\, dt \\ &= O(e^{-\epsilon x}) \qquad \text{as } x \to \infty. \end{aligned}$$

In both cases, F_2 is small beyond all orders in inverse powers of x.

End of lecture 4

3.3 Gamma function (again)

Γ can be analytically continued to $\mathbb{C}$, except at $z = 0, -1, -2, \ldots$, using

$$\Gamma(z) = \frac{\Gamma(z+1)}{z}.$$

Proof of Watson's lemma (continued). Recall that

$$F(x) = \int_0^\epsilon f(t)e^{-xt}\,dt + \int_\epsilon^T f(t)e^{-xt}\,dt = F_1(x) + F_2(x),$$

with $F_2(x) = O(e^{-\epsilon x})$ as $x \to \infty$. We want to show, for every N, that

$$F_1(x) = \sum_{n=0}^{N} a_n \frac{\Gamma(\alpha+\beta n+1)}{x^{\alpha+\beta n+1}} + o\Big(x^{-\alpha-\beta N-1}\Big).$$

Using property (d) of the Gamma function, consider

$$\begin{aligned}
&\left|F_1(x) - \sum_{n=0}^{N} a_n \frac{\Gamma(\alpha+\beta n+1)}{x^{\alpha+\beta n+1}}\right| \\
&= \left|\int_0^\epsilon f(t)e^{-xt}\,dt - \int_0^\infty \sum_{n=0}^{N} a_n t^{\alpha+\beta n} e^{-xt}\,dt\right| \\
&\le \int_0^\epsilon e^{-xt}\left|f(t) - \sum_{n=0}^{N} a_n t^{\alpha+\beta n}\right|\,dt + \left|\int_\epsilon^\infty e^{-xt}\sum_{n=0}^{N} a_n t^{\alpha+\beta n}\,dt\right| \\
&\equiv R_1 + R_2.
\end{aligned}$$

For R_2, put $u = t - \epsilon$. Then

$$\begin{aligned}
R_2 &\le e^{-\epsilon x}\sum_{n=0}^{N}|a_n|\int_\epsilon^\infty e^{-x(t-\epsilon)}t^{\alpha+\beta n}\,dt \\
&= e^{-\epsilon x}\sum_{n=0}^{N}|a_n|\int_0^\infty e^{-xu}(\epsilon+u)^{\alpha+\beta n}\,du.
\end{aligned}$$

For each n, there exist $C_n, k_n > 0$ such that

$$(\epsilon+u)^{\alpha+\beta n} < C_n e^{k_n u} \qquad \text{for all } u \in [0,\infty).$$

Hence, for $x > \max_{0\le n\le N} k_n$,

$$\begin{aligned}
R_2 &\le e^{-\epsilon x}\sum_{n=0}^{N} C_n|a_n|\int_0^\infty e^{(k_n-x)u}\,du \\
&= e^{-\epsilon x}\sum_{n=0}^{N}\frac{C_n|a_n|}{x-k_n} = O\left(\frac{e^{-\epsilon x}}{x}\right) \qquad \text{as } x\to\infty.
\end{aligned}$$

In R_1, use the asymptotic expansion of f. In particular,

$$f(t) - \sum_{n=0}^{N} a_n t^{\alpha+\beta n} = o(t^{\alpha+\beta N}) = O(t^{\alpha+\beta(N+1)}) \qquad \text{as } t\to 0^+,$$

where the last bound follows by taking one more term in the expansion. By the definition of O, for sufficiently small ϵ there is $K_3 > 0$ such that

$$\begin{aligned} R_1 &\le K_3 \int_0^\epsilon e^{-xt} t^{\alpha+\beta(N+1)}\, dt \\ &\le K_3 \int_0^\infty e^{-xt} t^{\alpha+\beta(N+1)}\, dt \\ &= K_3 \frac{\Gamma(\alpha+\beta(N+1)+1)}{x^{\alpha+\beta(N+1)+1}}. \end{aligned}$$

Since $\beta > 0$,

$$R_1 = o\Big(x^{-\alpha-\beta N-1}\Big) \qquad \text{as } x \to \infty.$$

This is true for every N, so

$$F_1(x) \sim \sum_{n=0}^{\infty} a_n \frac{\Gamma(\alpha+\beta n+1)}{x^{\alpha+\beta n+1}}.$$

Since F_2 is smaller than every inverse power of x, the same expansion holds for $F = F_1 + F_2$:

$$F(x) \sim \sum_{n=0}^{\infty} a_n \frac{\Gamma(\alpha+\beta n+1)}{x^{\alpha+\beta n+1}} \qquad \text{as } x \to \infty.$$

3.5 Laplace integrals and Laplace's method

Laplace integrals are of the form

$$I(x) = \int_a^b f(t) e^{x\phi(t)}\, dt,$$

where $b > a$, and f and ϕ are sufficiently nice functions. What happens to I as $x \to \infty$? Watson's lemma deals with the special case $\phi(t) = -t$, $a = 0$, $b = T$.

In general, we have Laplace's *principle of localisation*: the asymptotic expansion of I as $x \to \infty$ is determined by contributions from arbitrarily small regions around the maxima of ϕ. Figure 1 illustrates this principle.

(a) Suppose ϕ is monotonic on $[a, b]$. If $\phi'(t) > 0$ and $f(b) \neq 0$, let

$$u = \phi(t), \qquad t = \phi^{-1}(u).$$

Then

$$I(x) = \int_{\phi(a)}^{\phi(b)} \frac{f(t)}{\phi'(t)} e^{xu}\, du, \qquad t = \phi^{-1}(u).$$

Comparing with Example 2 from Section 3.2 gives

$$I(x) \sim \frac{f(b) e^{x\phi(b)}}{x\phi'(b)} \qquad \text{as } x \to \infty.$$

If $\phi'(t) < 0$ and $f(a) \neq 0$, then similarly

$$I(x) \sim \frac{f(a) e^{x\phi(a)}}{x|\phi'(a)|} \qquad \text{as } x \to \infty.$$

(b) Suppose $\phi(t)$ has one interior maximum at $t = c$. Assume

$$\phi'(c) = 0, \qquad \phi''(c) < 0, \qquad f(c) \neq 0.$$

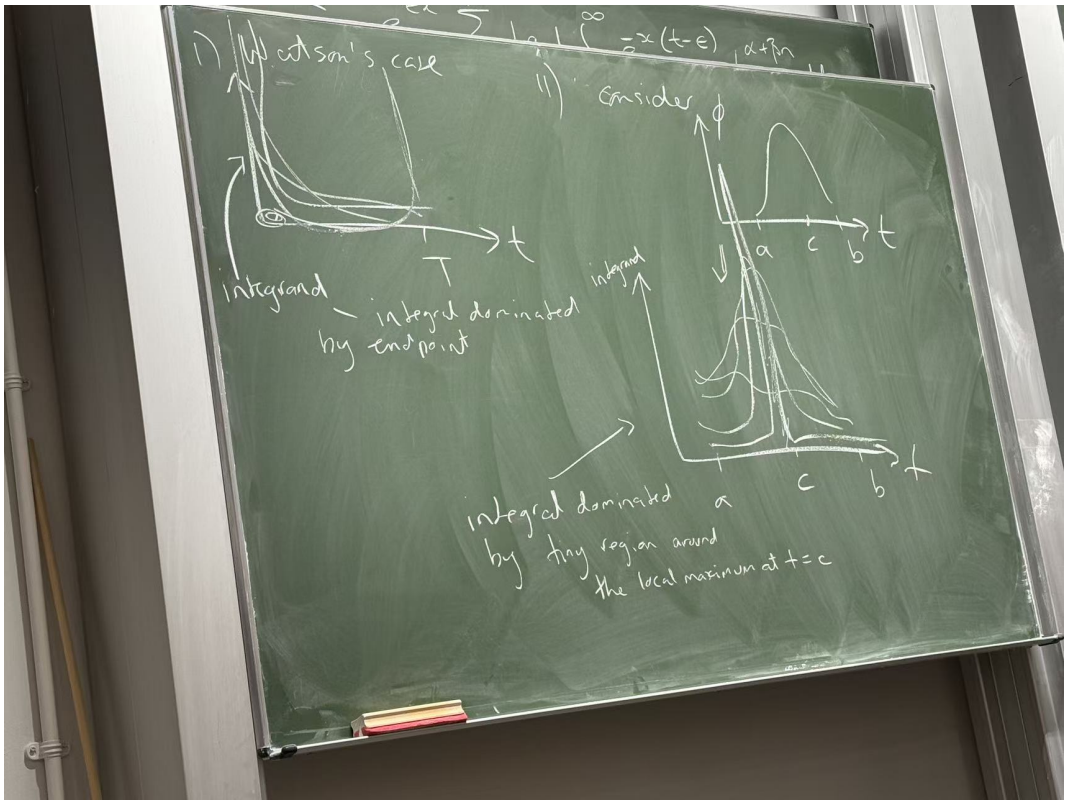


Figure 1: The three blackboard sketches of localisation. Left: Watson's case, where the integral is dominated by an endpoint. Upper right: a phase ϕ with an interior maximum at $t = c$. Lower right: the integrand becomes concentrated in a tiny region around c. Source: lecture blackboard photograph.

For small, positive ϵ, write

$$\Omega_\epsilon = [a, c-\epsilon] \cup [c+\epsilon, b]$$

and

$$\begin{aligned} I(x) &= \int_{c-\epsilon}^{c+\epsilon} f(t) e^{x\phi(t)}\, dt + \int_{\Omega_\epsilon} f(t) e^{x\phi(t)}\, dt \\ &\equiv I_1(x) + I_2(x). \end{aligned}$$

The small region around c is the region responsible for the concentration shown in Figure 1.

First, bound I_2. In the monotonic regions on either side of c,

$$\begin{aligned} I_2(x) &= \int_{c+\epsilon}^{b} f(t) e^{x\phi(t)}\, dt + \int_{a}^{c-\epsilon} f(t) e^{x\phi(t)}\, dt \\ &= O\left(\frac{e^{x\phi(c+\epsilon)}}{x} \right) + O\left(\frac{e^{x\phi(c-\epsilon)}}{x} \right). \end{aligned}$$

Here ϕ is monotonically decreasing on the right and monotonically increasing on the left. But

$$\phi(c+\epsilon) < \phi(c), \qquad \phi(c-\epsilon) < \phi(c),$$

so

$$I_2(x) = o\left(\frac{e^{x\phi(c)}}{x} \right) \qquad \text{as } x \to \infty.$$

Now consider I_1. Expand f and ϕ around $t = c$, assuming they are sufficiently differentiable:

$$\begin{aligned} \phi(t) &= \phi(c) + \underbrace{\phi'(c)(t-c)}_{=0} + \tfrac{1}{2}\phi''(c)(t-c)^2 + O((t-c)^3), \\ f(t) &= f(c) + O(t-c) \qquad \text{as } t \to c. \end{aligned}$$

Thus

$$\begin{aligned} I_1(x) &= \int_{c-\epsilon}^{c+\epsilon} [f(c) + \text{H.O.T.}] \exp\left[x\phi(c) + \tfrac{1}{2} x\phi''(c)(t-c)^2 + \text{H.O.T.} \right] dt \\ &= f(c) e^{x\phi(c)} \int_{c-\epsilon}^{c+\epsilon} e^{x\phi''(c)(t-c)^2/2} \left[1 + \text{H.O.T.} \right] dt. \end{aligned}$$

The idea is that the higher-order terms (H.O.T.) in powers of $(t-c)$ generate higher-order terms in an asymptotic series in inverse powers of x (see Chapling, Section 3.3.1, and B&O, Section 6.4). For now, we generate the leading-order term:

$$I_1(x) \sim f(c) e^{x\phi(c)} \int_{c-\epsilon}^{c+\epsilon} e^{x\phi''(c)(t-c)^2/2}\, dt \qquad \text{as } x \to \infty.$$

End of lecture 5

Laplace's method: an interior maximum (continued). In case (b), put

$$s = \sqrt{-x\phi''(c)}\,(t-c), \qquad A = \epsilon\sqrt{-x\phi''(c)}.$$

We use A for the quantity denoted by a on the blackboard, to distinguish it from the lower endpoint of the original integral. The leading local approximation becomes

$$I_1(x) \sim \frac{f(c) e^{x\phi(c)}}{\sqrt{-x\phi''(c)}} \int_{-A}^{A} e^{-s^2/2}\, ds.$$

Let $x \to \infty$, keeping $\epsilon > 0$ fixed. Then $A \to \infty$, and

$$\begin{aligned} I_1(x) &\sim \frac{f(c)e^{x\phi(c)}}{\sqrt{-x\phi''(c)}} \int_{-\infty}^{\infty} e^{-s^2/2}\, ds \\ &= f(c)e^{x\phi(c)} \left(\frac{2\pi}{-x\phi''(c)}\right)^{1/2}. \end{aligned}$$

Since $I_2 \ll I_1$ as $x \to \infty$, we obtain

$$\boxed{I(x) \sim f(c)e^{x\phi(c)} \left(\frac{2\pi}{-x\phi''(c)}\right)^{1/2}} \qquad \text{as } x \to \infty.$$

Error from extending the Gaussian limits. Recall that

$$\int_z^{\infty} e^{-t^2/2}\, dt \sim \frac{1}{z} e^{-z^2/2} \qquad \text{as } z \to \infty.$$

The blackboard records the error from replacing A by ∞ as

$$\text{error} = O\Big(x^{-1/2} e^{-\alpha x^2}\Big) \qquad \text{for some } \alpha > 0.$$

Transcription note (photograph 2): the displayed error is retained as written. With fixed ϵ, however, $A^2 = -\epsilon^2 x\phi''(c)$ is proportional to x. The omitted Gaussian tails, before multiplication by the outer prefactor, therefore have size $O(x^{-1/2}e^{-\alpha x})$, where $\alpha = -\epsilon^2\phi''(c)/2 > 0$. This concerns only the extension of the Gaussian limits; the higher-order Taylor terms give separate corrections.

For an alternative derivation, see Miller, Sections 3.3–3.4. For higher-order terms, see Chapling, Section 3.3.1, and B&O, Section 6.4.

Example 1. Consider

$$I(x) = \int_{-\infty}^{\infty} e^{-xt^2} e^{at}\, dt, \qquad x \to \infty,$$

where a is fixed. Take $f(t) = e^{at}$ and $\phi(t) = -t^2$. Then

$$c = 0, \quad \phi'(0) = 0, \quad \phi''(0) = -2, \quad f(0) = 1, \quad \phi(0) = 0.$$

Hence

$$I(x) \sim \sqrt{\frac{\pi}{x}}.$$

Indeed, completing the square gives the exact value

$$I(x) = e^{a^2/(4x)} \sqrt{\frac{\pi}{x}}.$$

Example 2. The integral written on the blackboard is

$$K_0(x) = \int_0^{\infty} e^{-t - x^2/(4t)}\, dt, \qquad x \to \infty.$$

Transcription note (photographs 5–6): both the symbol $K_0(x)$ and the measure dt are retained from the blackboard. This definition should be checked against the usual special-function notation; the calculation below applies to the integral displayed here.

One might try $f(t) = e^{-t}$ and $\phi(t) = -1/(4t)$, with large parameter x^2. But $\phi'(t) > 0$ for all $t > 0$, so this does not give an interior maximum of ϕ. Instead, consider the whole exponent

$$\psi(t) = -t - \frac{x^2}{4t}, \qquad \psi'(t) = -1 + \frac{x^2}{4t^2}.$$

Its maximum occurs at $t_{\max} = x/2$. This suggests rescaling by

$$xu = 2t, \qquad x\,du = 2\,dt.$$

Then

$$K_0(x) = \frac{x}{2}\int_0^\infty e^{x\phi(u)}\,du, \qquad \phi(u) = -\frac{u}{2} - \frac{1}{2u}.$$

Now

$$\phi'(u) = -\frac{1}{2} + \frac{1}{2u^2} = 0 \quad \Longrightarrow \quad u = 1 = c, \qquad \phi(1) = -1, \quad \phi''(1) = -1.$$

Therefore, for the integral as written,

$$K_0(x) \sim \frac{x}{2}e^{-x}\sqrt{\frac{2\pi}{x}} = \frac{\sqrt{2\pi x}}{2}e^{-x} \qquad \text{as } x \to \infty.$$

(c) Equal absolute maxima at an interior point and an endpoint. Suppose

$$\phi(b) = \phi(c) = \max_{t\in[a,b]} \phi(t), \qquad a < c < b,$$

with

$$\phi'(c) = 0, \quad \phi''(c) < 0, \quad \phi'(b) \neq 0, \qquad f(c) \neq 0, \quad f(b) \neq 0.$$

Figure 2 shows the situation. The nonstationary endpoint contributes a term of order $e^{x\phi(b)}/x$, as in case (a), whereas the interior maximum contributes a term of order $e^{x\phi(c)}/\sqrt{x}$, as in case (b). Since

$$\frac{e^{x\phi(b)}/x}{e^{x\phi(c)}/\sqrt{x}} = x^{-1/2} \to 0,$$

the interior point dominates the integral.

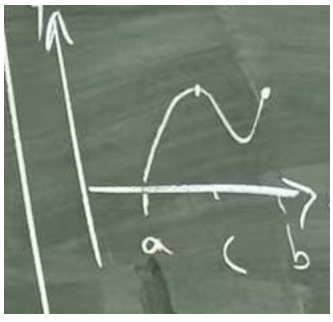


Figure 2: Equal maxima at an interior stationary point c and a nonstationary endpoint b. Source: direct crop of supplied blackboard photograph 6, archived as `source-06.jpg`.

(d) Several interior local maxima. Suppose the interior local maxima c_i, $i = 1, \ldots, n$, satisfy

$$\phi'(c_i) = 0, \qquad \phi''(c_i) < 0, \qquad f(c_i) \neq 0.$$

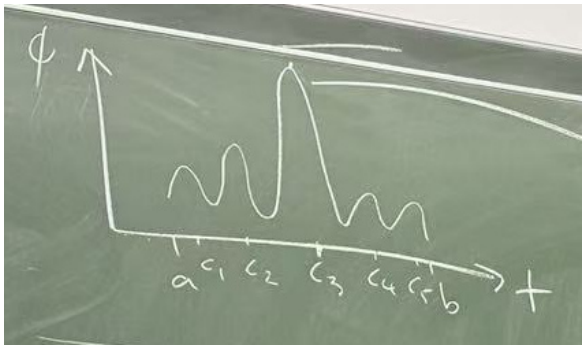
Only the absolute maxima matter at leading order. For instance, if $\phi(c_3) > \phi(c_i)$ for $i = 1, 2, 4, 5$, then

$$\frac{e^{x\phi(c_3)}}{\sqrt{x}} \gg \frac{e^{x\phi(c_i)}}{\sqrt{x}} \qquad \text{as } x \to \infty.$$

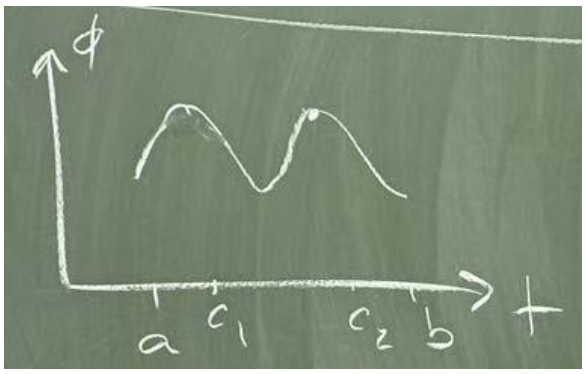
This is illustrated in Figure 3(a). If instead two maxima have equal height, $\phi(c_1) = \phi(c_2) = \max \phi$, both contributions must be included:

$$I(x) \sim e^{x\phi(c_1)} \sqrt{\frac{2\pi}{-x\phi''(c_1)}} + e^{x\phi(c_2)} \sqrt{\frac{2\pi}{-x\phi''(c_2)}}.$$

Transcription note (photograph 8): this is the formula as written on the blackboard. For general f, the two terms must also be multiplied by $f(c_1)$ and $f(c_2)$, respectively; the displayed formula applies when $f \equiv 1$. If the leading coefficients cancel, further terms are needed before asserting an asymptotic equivalence.



(a) One highest maximum.



(b) Two equal maxima.

Figure 3: Several interior maxima: lower maxima are exponentially smaller, while equal absolute maxima contribute together. Source: direct crops of supplied blackboard photograph 8, archived as `source-08.jpg`.

(e) A stationary endpoint maximum. Suppose the endpoint b is a local and absolute maximum, with

$$\phi'(b) = 0, \qquad \phi''(b) < 0, \qquad f(b) \neq 0.$$

Proceed as in case (b), but the local integral now covers only one side of the maximum:

$$\begin{aligned} I_1(x) &= \int_{b-\epsilon}^{b} f(t) e^{x\phi(t)}\, dt \\ &\sim \frac{f(b) e^{x\phi(b)}}{\sqrt{-x\phi''(b)}} \int_{-\infty}^{0} e^{-s^2/2}\, ds \\ &= \frac{1}{2} f(b) e^{x\phi(b)} \sqrt{\frac{2\pi}{-x\phi''(b)}} \qquad \text{as } x \to \infty. \end{aligned}$$

Thus a stationary endpoint gives half the corresponding full Gaussian contribution. The same conclusion holds at a stationary left endpoint.

Example. Consider

$$I(x) = \int_0^{3\pi/2} e^{x \sin^2 t}\, dt, \qquad f(t) = 1, \quad \phi(t) = \sin^2 t.$$

The absolute maxima are at $t = \pi/2$ (interior) and $t = 3\pi/2$ (endpoint), as shown in Figure 4. At both points,

$$\phi = 1, \qquad \phi' = 0, \qquad \phi'' = 2\cos(2t) = -2.$$

Hence

$$I(x) \sim \frac{e^x}{\sqrt{2x}}\sqrt{2\pi} + \frac{1}{2}\frac{e^x}{\sqrt{2x}}\sqrt{2\pi}$$
$$= \frac{3}{2}\sqrt{\frac{\pi}{x}}\,e^x \qquad \text{as } x \to \infty.$$



Figure 4: The phase $\phi(t) = \sin^2 t$ on $[0, 3\pi/2]$, with one interior maximum and one stationary endpoint maximum. Source: direct crop of supplied blackboard photograph 9, archived as `source-09.jpg`.

(f) A higher-order interior maximum. Suppose c is the dominant interior maximum and

$$\phi'(c) = \phi''(c) = \cdots = \phi^{(p-1)}(c) = 0, \qquad \phi^{(p)}(c) < 0,$$

where p is an even positive integer, and $f(c) \neq 0$. Proceed as in case (b), now using

$$\phi(t) = \phi(c) + \frac{\phi^{(p)}(c)}{p!}(t-c)^p + \cdots.$$

Then

$$I(x) \sim f(c)e^{x\phi(c)} \int_{c-\epsilon}^{c+\epsilon} \exp\left[\frac{x\phi^{(p)}(c)}{p!}(t-c)^p\right] dt.$$

Use the rescaling

$$s = \left[\frac{-x\phi^{(p)}(c)}{p!}\right]^{1/p} (t-c).$$

As $x \to \infty$, this gives

$$I(x) \sim f(c)e^{x\phi(c)} \left[\frac{p!}{-x\phi^{(p)}(c)}\right]^{1/p} \int_{-\infty}^{\infty} e^{-s^p}\, ds.$$

To evaluate the remaining integral, put $u = s^p$ on $s \geq 0$:

$$\int_0^\infty e^{-s^p}\, ds = \frac{1}{p}\int_0^\infty e^{-u}u^{1/p-1}\, du$$
$$= \frac{1}{p}\Gamma\left(\frac{1}{p}\right) = \Gamma\left(1+\frac{1}{p}\right).$$

Since p is even, the full integral is twice this value. Therefore

$$\boxed{I(x) \sim 2f(c)e^{x\phi(c)} \left[\frac{p!}{-x\phi^{(p)}(c)}\right]^{1/p} \Gamma\left(1+\frac{1}{p}\right)} \qquad \text{as } x \to \infty.$$

In particular, the algebraic factor is $x^{-1/p}$, so the contribution is proportional to $x^{-1/p}e^{x\phi(c)}$.

End of lecture 6

Laplace's method (continued): higher-order terms

(g) How do we obtain higher-order terms? For

$$I(x) = \int_a^b f(t) e^{x\phi(t)}\, dt,$$

retain more terms in the Taylor expansions of both f and ϕ about the absolute maximum $t = c$. For general f and ϕ the algebra becomes lengthy; see Section 3.4.4 of Chapling for the general treatment. If $f(c) = 0$, higher-order terms may be needed even to find the leading nonzero contribution.

Example 1. Consider

$$J(x) = \int_0^\pi e^{x \sin t}\, dt, \qquad x \to \infty.$$

Here $f = 1$ and $\phi(t) = \sin t$, whose unique absolute maximum is the interior point $c = \pi/2$. Since $\phi(c) = 1$ and $\phi''(c) = -1$,

$$J(x) \sim \sqrt{\frac{2\pi}{x}}\, e^x.$$

To obtain the next term, localise to $|t - \pi/2| < \epsilon$ and put $u = t - \pi/2$. Then $\sin t = \cos u$, and the local integral is

$$e^x \int_{-\epsilon}^{\epsilon} e^{-xu^2/2} \exp\!\left(\frac{xu^4}{4!} - \frac{xu^6}{6!} + \cdots \right) du.$$

With $s = \sqrt{x}\, u$, this becomes

$$\frac{e^x}{\sqrt{x}} \int_{-\epsilon\sqrt{x}}^{\epsilon\sqrt{x}} e^{-s^2/2} \exp\!\left(\frac{s^4}{24x} - \frac{s^6}{720x^2} + \cdots \right) ds.$$

Expanding the second exponential and extending the Gaussian integrals to the whole real line gives

$$\begin{aligned} J(x) &= \frac{e^x}{\sqrt{x}} \left[\int_{-\infty}^{\infty} e^{-s^2/2}\, ds + \frac{1}{24x} \int_{-\infty}^{\infty} s^4 e^{-s^2/2}\, ds + O(x^{-2}) \right] \\ &= \boxed{\sqrt{\frac{2\pi}{x}}\, e^x \left[1 + \frac{1}{8x} + O(x^{-2}) \right]}. \end{aligned}$$

We used

$$\int_{-\infty}^{\infty} e^{-s^2/2}\, ds = \sqrt{2\pi}, \qquad \int_{-\infty}^{\infty} s^4 e^{-s^2/2}\, ds = 3\sqrt{2\pi}.$$

Exercise. Bound the next largest term and hence justify the asymptotic expansion. The exponential expansion is not uniform on the full expanding interval: one can first restrict to $|s| \le x^\delta$ with $0 < \delta < 1/4$, bound the integrated Taylor remainder by $O(x^{-2})$, and estimate the remaining Gaussian tails separately. The contribution outside the original neighbourhood of c is exponentially smaller.

Example 2. A stationary endpoint. Consider

$$I(x) = \int_x^\infty e^{-t - x^2/t}\, dt, \qquad x \to \infty.$$

As in the earlier example involving $K_0(x)$, use a co-moving variable $t = xu$. Then

$$I(x) = x \int_1^\infty e^{x\phi(u)}\, du, \qquad \phi(u) = -u - u^{-1}.$$

The maximum is at the endpoint $u = 1$, where

$$\phi(1) = -2, \quad \phi'(1) = 0, \quad \phi''(1) = -2, \quad \phi'''(1) = 6.$$

Writing $v = u - 1$, we have

$$\phi(1 + v) = -2 - v^2 + v^3 - v^4 + \cdots .$$

Localise to $1 \le u \le 1 + \epsilon$, and set $s = (u - 1)\sqrt{x}$. The local integral becomes

$$\sqrt{x}\, e^{-2x} \int_0^{\epsilon\sqrt{x}} e^{-s^2} \exp\left(\frac{s^3}{\sqrt{x}} - \frac{s^4}{x} + \cdots \right) ds.$$

Expanding and integrating term by term, with the remainder controlled after integration, gives

$$\begin{aligned} I(x) &= \sqrt{x}\, e^{-2x} \left[\int_0^\infty e^{-s^2}\, ds + x^{-1/2} \int_0^\infty s^3 e^{-s^2}\, ds + O(x^{-1}) \right] \\ &= \sqrt{x}\, e^{-2x} \left[\frac{\sqrt{\pi}}{2} + \frac{1}{2\sqrt{x}} + O(x^{-1}) \right] \\ &= \boxed{\frac{\sqrt{\pi x}}{2} e^{-2x} \left[1 + \frac{1}{\sqrt{\pi x}} + O(x^{-1}) \right]}. \end{aligned}$$

The useful Gaussian moment identity is

$$\int_0^\infty t^{2n+1} e^{-t^2/a^2}\, dt = \frac{n!\, a^{2n+2}}{2}, \qquad a > 0, \quad n = 0, 1, 2, \ldots .$$

Unlike at a symmetric interior maximum, the odd cubic correction does not integrate to zero on the half-line. Thus the first relative correction has order $x^{-1/2}$.

3.6 Oscillatory integrals and the method of stationary phase

Consider the Fourier-type integral

$$F(x) = \int_a^b f(t) e^{ix\phi(t)}\, dt, \qquad x \to \infty,$$

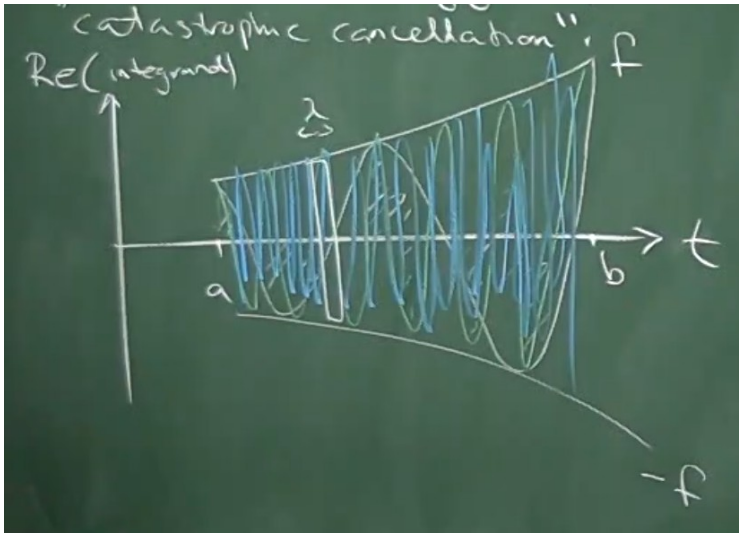
where ϕ is real. The function f is the *amplitude* or *envelope*, $x\phi(t)$ is the *phase*, and

$$k(t) = x\phi'(t)$$

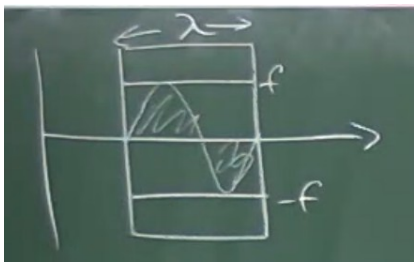
is the local wavenumber (or angular frequency). Wherever $\phi'(t) \neq 0$, the local wavelength is

$$\lambda(t) = \frac{2\pi}{|x\phi'(t)|}.$$

Although $|e^{ix\phi(t)}| = 1$, the integrand oscillates increasingly rapidly as $x \to \infty$. Away from stationary points, λ becomes much smaller than the lengthscale on which f varies. Locally f then looks nearly constant over a wavelength, so positive and negative contributions almost cancel.



(a) Rapid oscillations within the envelope.



(b) Cancellation over one wavelength.

Figure 5: Oscillatory cancellation. Direct crops of supplied blackboard photographs 7 and 8, archived as `source-07.png` and `source-08.png`.

(a) The Riemann–Lebesgue lemma

For the linear phase $\phi(t) = t$, if

$$\int_a^b |f(t)|\,dt < \infty, \qquad -\infty \le a < b \le \infty,$$

then

$$\boxed{F(x) = \int_a^b f(t)e^{ixt}\,dt \longrightarrow 0 \quad \text{as } |x| \to \infty.}$$

See Appendix A of Chapling for a proof. The lemma gives convergence to zero, but by itself does not specify a decay rate.

A first decay estimate. Suppose initially that a, b are finite, f is absolutely continuous on $[a, b]$, $f' \in L^1(a, b)$, and $f(a), f(b) \neq 0$. Integration by parts gives

$$\begin{aligned} F(x) &= \left[\frac{f(t)e^{ixt}}{ix}\right]_a^b - \frac{1}{ix}\int_a^b f'(t)e^{ixt}\,dt \\ &= \frac{1}{ix}\left[f(b)e^{ixb} - f(a)e^{ixa} - \int_a^b f'(t)e^{ixt}\,dt\right]. \end{aligned}$$

By the Riemann–Lebesgue lemma applied to f', the last integral tends to zero. Therefore

$$\boxed{F(x) = \frac{f(b)e^{ixb} - f(a)e^{ixa}}{ix} + o(x^{-1}) = O(x^{-1}).}$$

The leading contribution is controlled by the boundary values of f: cancellation is imperfect at the endpoints. An additive remainder is used here because the oscillatory boundary term can vanish for some values of x, so a ratio statement using $\sim$ need not hold.

Vanishing boundary terms and faster decay. Repeated integration by parts gives, under the corresponding regularity and integrability assumptions,

$$F(x) = \sum_{k=0}^{N-1} \frac{(-1)^k}{(ix)^{k+1}}\left[f^{(k)}(t)e^{ixt}\right]_a^b + \frac{(-1)^N}{(ix)^N}\int_a^b f^{(N)}(t)e^{ixt}\,dt.$$

Thus faster decay depends on vanishing boundary values of successive derivatives. In particular, if f is smooth on a finite interval and *all* of its derivatives vanish at both endpoints, then

$$F(x) = O(x^{-N}) \qquad \text{for every } N \in \mathbb{Z}_{>0}.$$

This is decay beyond all algebraic orders in x^{-1}.

For the whole-line Fourier integral,

$$F(x) = \int_{-\infty}^{\infty} f(t)e^{ixt}\,dt,$$

the same conclusion holds, for example, if $f \in C^\infty(\mathbb{R})$, each derivative $f^{(k)}$ is integrable, and $f^{(k)}(t) \to 0$ as $t \to \pm\infty$ for every $k \geq 0$:

$$|F(x)| \leq |x|^{-N} \int_{-\infty}^{\infty} |f^{(N)}(t)|\,dt.$$

Qualification of the blackboard statement: $f(a) = f(b) = 0$ alone does not imply decay beyond all orders. Likewise, smoothness and $f(t) \to 0$ at infinity alone do not guarantee this rate; the derivatives must allow the repeated integrations by parts and control the remainders.

End of lecture 7

Oscillatory integrals (continued)

We continue to study

$$F(x) = \int_a^b f(t)e^{ix\phi(t)}\,dt, \qquad x \to +\infty,$$

with real phase ϕ. The signed local wavenumber is $x\phi'(t)$, while the local wavelength is $2\pi/|x\phi'(t)|$. For the linear phase, repeated integration by parts gives decay beyond all algebraic orders when all boundary terms vanish and the derivatives are integrable, as described above.

Remark on Fourier decay. The board mentions the Paley–Wiener theorem in connection with how rapidly a Fourier transform can decay. Such restrictions require precise hypotheses, for example on support and analytic continuation; the whole-line integral alone imposes no universal prohibition on Gaussian decay. The photograph does not specify enough hypotheses to state the intended theorem (see archived photograph `source-01.jpg`).

(b) A non-stationary, monotone phase

Suppose a, b are finite, $\phi'(t) \neq 0$ on $[a, b]$, and f and ϕ have sufficient regularity for the following change of variables and integration by parts. Put

$$u = \phi(t), \qquad t = \phi^{-1}(u).$$

Then, with oriented limits also when ϕ is decreasing,

$$F(x) = \int_{\phi(a)}^{\phi(b)} \frac{f(\phi^{-1}(u))}{\phi'(\phi^{-1}(u))} e^{ixu}\,du.$$

Equivalently, integration by parts directly in t gives

$$F(x) = \frac{1}{ix}\left[\frac{f(t)}{\phi'(t)}e^{ix\phi(t)}\right]_a^b - \frac{1}{ix}\int_a^b \left(\frac{f(t)}{\phi'(t)}\right)' e^{ix\phi(t)}\,dt.$$

If $(f/\phi')'$ is integrable, the transformed remainder tends to zero by the Riemann–Lebesgue lemma. Hence

$$\boxed{F(x) = \frac{1}{ix}\left[\frac{f(t)}{\phi'(t)}e^{ix\phi(t)}\right]_a^b + o(x^{-1}) = O(x^{-1}).}$$

When $f(a), f(b) \neq 0$, the endpoints supply the possible leading term. An additive expansion is used because this boundary term may have zeros.

(c) A stationary point: localisation

Suppose now that there is one interior stationary point $c \in (a, b)$:

$$\phi'(c) = 0, \qquad \phi'(t) \neq 0 \quad (t \neq c).$$

Near c, the local wavelength grows rather than shrinking to zero; oscillatory cancellation is less pronounced. A relatively flat *plateau* in the phase makes this neighbourhood dominate the integral, although its contribution still tends to zero.

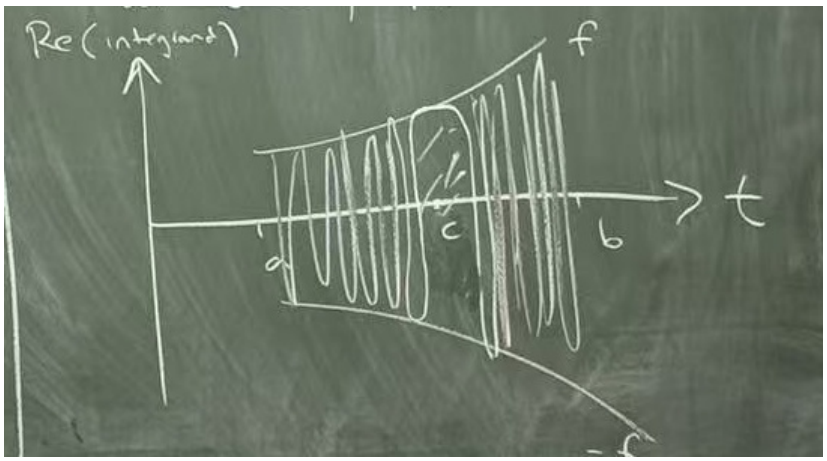


Figure 6: Reduced oscillatory cancellation near $t = c$. Direct crop of supplied photograph 2, archived as `source-02.jpg`.

Localisation principle (Stokes and Kelvin). The asymptotic expansion of an oscillatory integral is controlled by points of stationary phase and possibly the endpoints. For small fixed $\epsilon > 0$, split

$$\begin{aligned} F(x) &= \int_a^{c-\epsilon} f(t)e^{ix\phi(t)}\,dt + \int_{c-\epsilon}^{c+\epsilon} f(t)e^{ix\phi(t)}\,dt + \int_{c+\epsilon}^{b} f(t)e^{ix\phi(t)}\,dt \\ &= \int_{c-\epsilon}^{c+\epsilon} f(t)e^{ix\phi(t)}\,dt + O(x^{-1}). \end{aligned}$$

The two outer integrals have non-stationary phase and are $O(x^{-1})$ under the preceding regularity assumptions.

A non-degenerate stationary point

Assume in addition that $\phi''(c) \neq 0$ and $f(c) \neq 0$. Taylor expansion near c gives

$$\phi(t) = \phi(c) + \frac{\phi''(c)}{2}(t-c)^2 + \cdots, \qquad f(t) = f(c) + \cdots.$$

Thus the leading local contribution is obtained from

$$f(c)e^{ix\phi(c)}\int_{c-\epsilon}^{c+\epsilon}\exp\left(\frac{ix\phi''(c)}{2}(t-c)^2\right)dt.$$

Introduce

$$s=\sqrt{\frac{x|\phi''(c)|}{2}}(t-c),\qquad \sigma=\operatorname{sgn}(\phi''(c))\in\{-1,1\},\qquad L=\epsilon\sqrt{\frac{x|\phi''(c)|}{2}}.$$

The quadratic model becomes

$$f(c)e^{ix\phi(c)}\sqrt{\frac{2}{x|\phi''(c)|}}\int_{-L}^{L}e^{i\sigma s^2}\,ds.$$

As $L\to\infty$, use the Fresnel integral (also in example sheets 1 and 2)

$$\int_{-\infty}^{\infty}e^{i\sigma s^2}\,ds=\sqrt{\pi}\,e^{i\sigma\pi/4}.$$

This integral is conditionally convergent. The quadratic replacement is a local asymptotic argument, with the non-stationary part estimated separately; it is not a uniform Taylor approximation on the full expanding s-interval. The stationary-phase formula is

$$\boxed{F(x)=f(c)\sqrt{\frac{2\pi}{x|\phi''(c)|}}\exp\left(ix\phi(c)+\frac{i\pi}{4}\operatorname{sgn}(\phi''(c))\right)+o(x^{-1/2}).}$$

For this single nonzero contribution one may equivalently write $\sim$. The board's compact notation $f(c)e^{ix\phi(c)}\sqrt{2\pi/(-ix\phi''(c))}$ uses the square-root branch giving the phase factor above. The contribution is $O(x^{-1/2})$, larger than the non-stationary endpoint contributions of order x^{-1}.

Example: the large-argument behaviour of J_0

Use the integral representation

$$J_0(r)=\frac{1}{2\pi}\int_0^{2\pi}e^{ir\sin t}\,dt,\qquad r\to+\infty.$$

Here $f=1/(2\pi)$, $\phi(t)=\sin t$, and there are two stationary points:

c	$\phi(c)$	$\phi''(c)$	leading contribution
$\pi/2$	1	-1	$(2\pi r)^{-1/2}e^{i(r-\pi/4)}$
$3\pi/2$	-1	1	$(2\pi r)^{-1/2}e^{-i(r-\pi/4)}$

Adding their contributions gives

$$\boxed{J_0(r)=\sqrt{\frac{2}{\pi r}}\cos\left(r-\frac{\pi}{4}\right)+o(r^{-1/2}).}$$

The board writes this informally using $\sim$. Because the cosine has infinitely many zeros, a global ratio interpretation is inappropriate; the additive form specifies the asymptotic approximation.

Final points on stationary phase.

- Each non-degenerate stationary point with nonzero amplitude supplies a contribution at order $x^{-1/2}$; these contributions must be added and may cancel.
- Non-stationary endpoints can contribute at order x^{-1}. These terms are algebraically small, rather than exponentially small as away from an isolated maximum in Laplace's method.
- Use care with $\sim$ for oscillatory expressions with zeros.
- Laplace's method is often easier for controlling higher-order terms. This motivates deforming oscillatory integrals into a form with exponential decay.

3.7 Method of steepest descent

Consider

$$I(x) = \int_C f(z) e^{x\phi(z)}\, dz, \qquad x \to +\infty,$$

where C is a contour in $\mathbb{C}$ and f, ϕ are analytic in a domain containing the contour and any allowed deformation of it. Such integrals arise in special functions and physical applications. Oscillatory integrals can also be put in this form, allowing higher-order terms to be calculated using Laplace-type reasoning.

Write

$$z = p + iq, \qquad \phi(z) = u(p,q) + iv(p,q), \qquad p, q, u, v \in \mathbb{R}.$$

Then

$$I(x) = \int_C f(z) \underbrace{e^{xu(p,q)}}_{\text{amplitude factor}} \underbrace{e^{ixv(p,q)}}_{\text{oscillatory factor}} dz.$$

The real part u controls exponential growth or decay, while the imaginary part v controls oscillation. On a curve of constant v, the phase factor is constant, suggesting the use of Laplace's method.

Interlude: the Cauchy–Riemann equations

Analyticity of ϕ gives

$$u_p = v_q, \qquad u_q = -v_p.$$

In the (p,q)-plane,

$$\nabla u = (u_p, u_q), \qquad \nabla v = (v_p, v_q), \qquad \nabla u \cdot \nabla v = 0.$$

Thus the regular level curves of u and v are orthogonal. Moreover, ∇v is tangent to curves of constant u, and ∇u is tangent to curves of constant v. Along a constant-v curve oriented in the direction $-\nabla u$, the real part u decreases most rapidly per unit arclength:

$$\frac{du}{ds} = -|\nabla u|.$$

The opposite orientation gives steepest ascent. These statements apply at regular points; at a stationary point the gradients vanish and the local geometry needs separate examination.

Example: $\phi(z) = z^2$

Here

$$u = p^2 - q^2, \qquad v = 2pq.$$

The constant-u and constant-v curves are orthogonal families of hyperbolas away from the stationary point at $z = 0$; the zero levels include straight lines. The oriented red constant-v curves in Figure 7 follow decreasing u, from the regions near the real axis towards the regions near the imaginary axis.

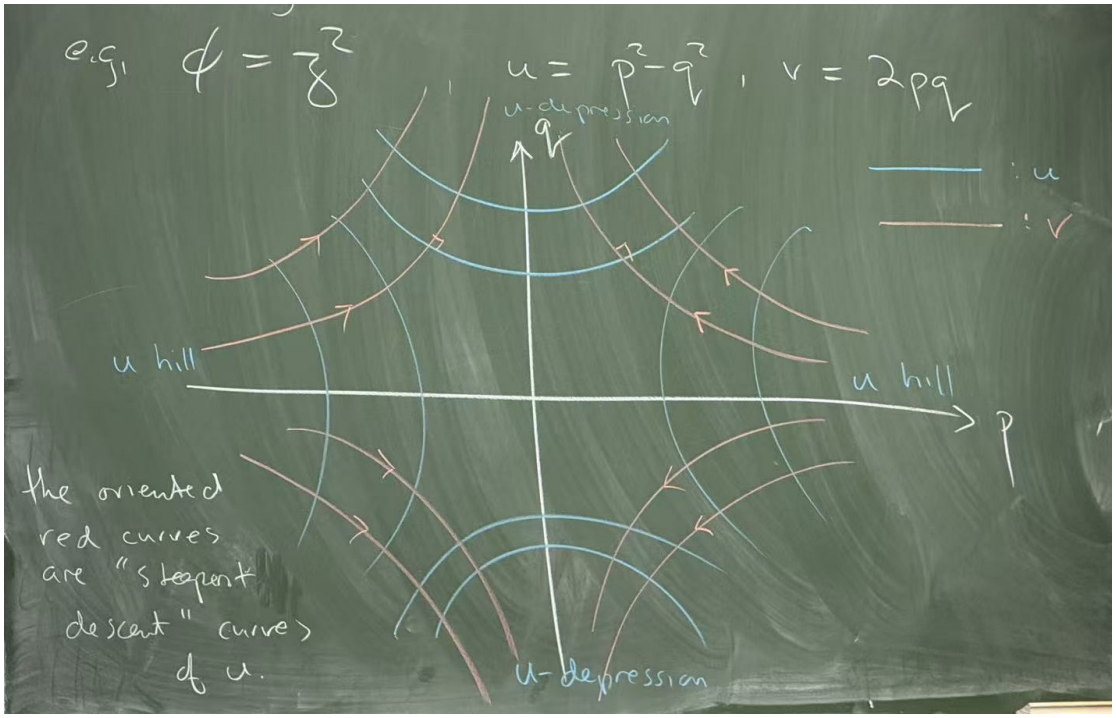


Figure 7: Level curves for $\phi(z) = z^2$: blue curves have constant $u = p^2 - q^2$, and red curves have constant $v = 2pq$. Red arrows mark steepest descent of u. Direct crop of supplied photograph 10, archived as `source-10.jpg`.

Along the imaginary axis $z = iq$, $u = -q^2$ decreases away from the origin; along the real axis $z = p$, $u = p^2$ increases away from it. Thus the origin is a saddle of u, with descent directions along the imaginary axis and ascent directions along the real axis.

Source photographs. All ten supplied blackboard photographs are preserved in the directory

`figures/asymptotic-lecture8/`

They are named `source-01.jpg`–`source-10.jpg`, in the supplied order. Photographs 1–6 cover oscillatory integrals and stationary phase; photographs 7–10 introduce steepest descent and its geometry. The embedded figures are crops of these originals.

End of lecture 8

3.7 Method of steepest descent (contd)

$$I(x) = \int_C f(z) e^{x\phi(z)}\, dz, \qquad \text{what does } I \text{ do as } x \to \infty.$$

C a contour in $\mathbb{C}$ & f & ϕ analytic in a domain containing C.

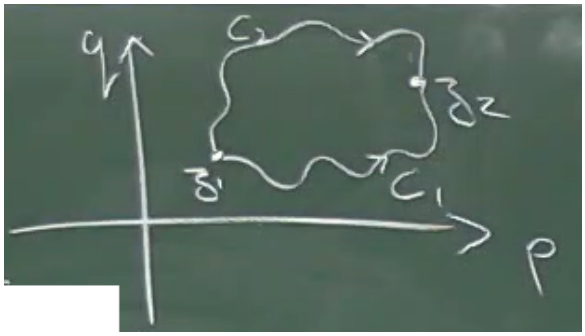
$$\phi = u + iv.$$

Interlude 2: Cauchy's theorem

$$\oint_C f e^{\phi x}\, dz = 0$$

if f & ϕ analytic in the closed domain bounded by C.

In particular



$$\int_{C_1} f e^{\phi x}\, dz = \int_{C_2} f e^{\phi x}\, dz.$$

Steepest descent method: cases

(a) Suppose $\nabla u \neq 0$ on the domain.

(i) Suppose $\mathrm{Im}(\phi) = v$ is constant on C.
Let $v = \mathrm{cst} = v_0 \in \mathbb{R}$ on C.

$\Rightarrow$ C is a level curve of v & hence a steepest descent/ascent curve of u.

- Parametrize C s.t.

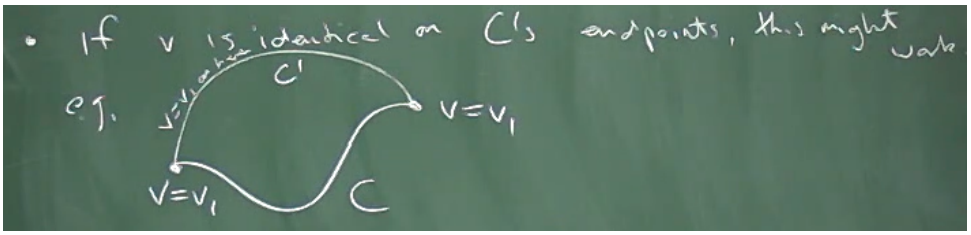
$$C = \big\{ z \in \mathbb{C}; \quad z(t) = p(t) + iq(t) \ \&\ a \leq t \leq b \big\}.$$

$$I(x) = e^{ixv_0} \int_a^b e^{xu}\, \mathrm{Re}\left(f \frac{dz}{dt} \right) dt + ie^{ixv_0} \int_a^b e^{xu}\, \mathrm{Im}\left(f \frac{dz}{dt} \right) dt.$$

(ii) Suppose v is not constant on C.

- Method: deform C, using Cauchy's thm, to a new curve C' upon which v is constant.
- If v is identical on C's endpoints, this might work.

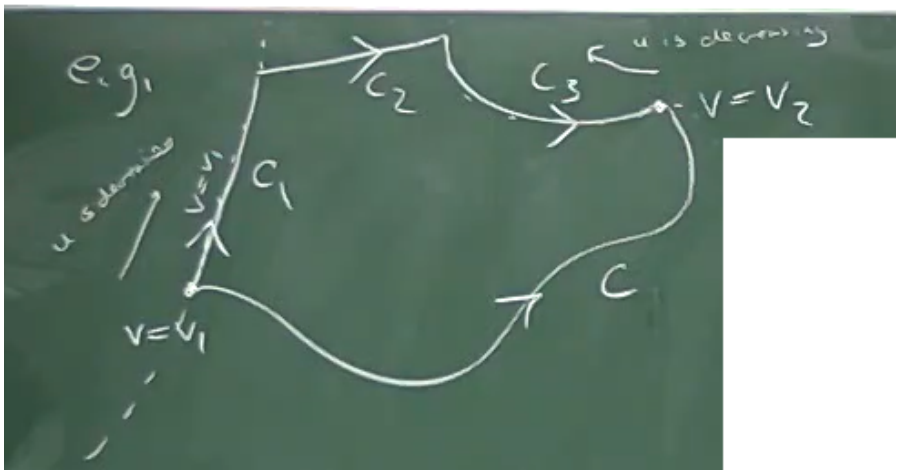
e.g.



- Not always possible, & certainly not if v takes different values at the 2 endpoints.

If v takes different values, we deform C to a curve that follows constant-v curves with bridging curves between them. And then show that the integrals on bridging curves don't contribute much to the leading order asymptotics.

e.g.



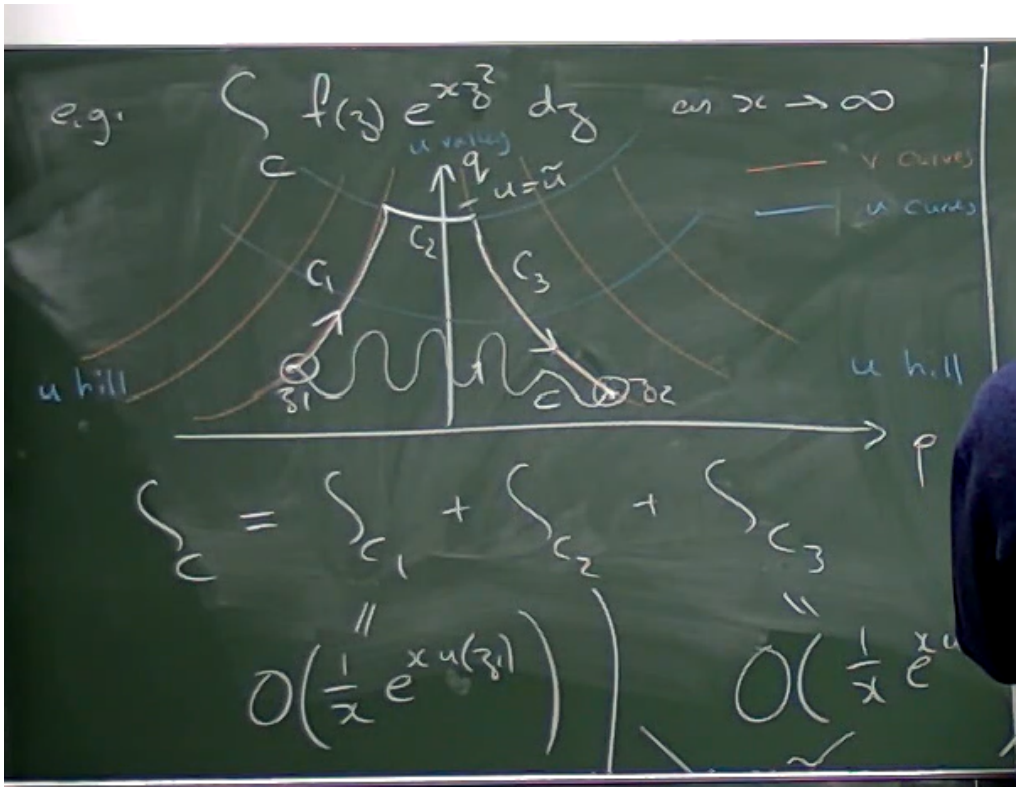
$$v_1 \neq v_2, \qquad \int_C = \int_{C_1} + \underbrace{\int_{C_2}}_{\substack{\text{Want to show this} \\ \text{is negligible!}}} + \int_{C_3}.$$

How to select which direction to go along constant-v curves?

u is monotonically increasing or decreasing along constant-v curves we select direction we traverse so that u is decreasing away from the endpoints.

e.g.

$$\int_C f(z)e^{xz^2}\,dz \qquad \text{as } x \to \infty.$$



$$\int_C = \int_{C_1} + \int_{C_2} + \int_{C_3}, \qquad \int_{C_1} = O\left(\frac{1}{x} e^{xu(z_1)}\right).$$

e.g.

$$J_0(r) = \frac{1}{\pi}\int_0^\pi e^{ir\cos t}\,dt \qquad \text{as } r\to\infty \qquad \left(\sim \sqrt{\frac{2}{\pi r}}\cos\Big(r-\frac{\pi}{4}\Big)\right).$$

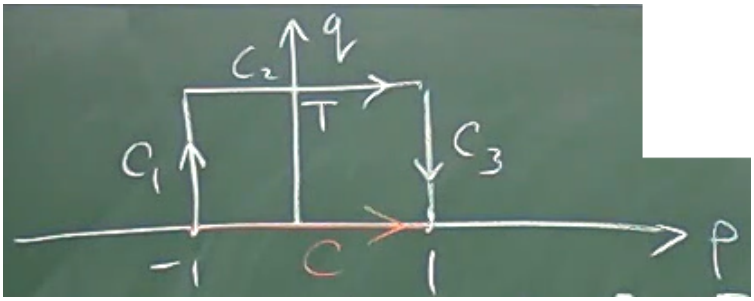
$$s = \cos t.$$

$$J_0 = \frac{1}{\pi}\int_{-1}^{1}\frac{1}{\sqrt{1-s^2}}e^{irs}\,ds.$$

Let $s = z \in \mathbb{C}$.

$$\phi = iz \quad \Rightarrow \quad \begin{matrix} u = -q, \\ v = p. \end{matrix}$$

Constant-v curves are $p = \text{cst}$.



$$\pi J_0 = \int_{C_1} + \int_{C_2} + \int_{C_3}.$$

$$\begin{aligned} C_1: &\quad z = -1+it, &\quad t\in[0,T],\\ C_2: &\quad z = t+iT, &\quad t\in[-1,1],\\ C_3: &\quad z = 1-it, &\quad t\in[-T,0]. \end{aligned}$$

$$\begin{aligned} \int_{C_1} &= i\int_0^T \frac{e^{ir(-1+it)}}{\sqrt{t(t+2i)}}\,dt \\ &= ie^{-ir}\int_0^T \frac{e^{-rt}}{\sqrt{t(t+2i)}}\,dt, \end{aligned}$$

$$\begin{aligned} \int_{C_3} &= -i\int_{-T}^0 \frac{e^{ir(1-it)}}{\sqrt{t(t+2i)}}\,dt \qquad (t\to -t) \\ &= -ie^{ir}\int_0^T e^{-rt}\big(t(t-2i)\big)^{-1/2}\,dt. \end{aligned}$$

$$\int_{C_2} = \int_{-1}^1 \frac{e^{irt-rT}}{\sqrt{1-(t+iT)^2}}\,dt.$$

$$\left|\int_{C_2}\right| \le e^{-rT}\int_{-1}^1 \left|\frac{1}{\sqrt{1-(t+iT)^2}}\right| dt = O\big(e^{-rT}\big).$$

$$J_0 \sim \frac{2}{\pi}\,\mathrm{Re}\left[e^{ir-i\pi/2}\int_0^T e^{-rt}\frac{1}{\sqrt{t(t-2i)}}\,dt\right] \qquad \text{as } r\to\infty.$$

Note

$$\frac{1}{\sqrt{t-2i}} = \sum_{n=0}^{\infty} a_n t^n, \qquad a_n = \cdots .$$

Watson's lemma applies, $\alpha = -\frac{1}{2}$, $\beta = 1$.

$$\begin{aligned} J_0 &\sim \frac{2}{\pi} \operatorname{Re}\left[e^{ir - i\pi/2} r^{-1/2} \sum_{n=0}^{\infty} \frac{a_n \Gamma(\frac{1}{2}+n)}{r^n} \right] \qquad \text{as } r \to \infty \\ &\sim \sqrt{\frac{2}{\pi r}} \left[\cos\Big(r - \frac{\pi}{4}\Big) + \frac{1}{8r} \sin\Big(r - \frac{\pi}{4}\Big) \right. \\ &\qquad \left. - \frac{9}{128 r^2} \cos\Big(r - \frac{\pi}{4}\Big) + \cdots \right]. \end{aligned}$$

Summary of procedure

- Deform C to C'.
- Choose C' so that it follows constant-v curves as much as possible (using bridging curves if needed).
- Choose orientation of C' so that it is a steepest descent curve of u, away from endpoints.
- Show that contributions from bridging curves don't contribute to the asymptotics of the integral.
- Use Watson's lemma or endpoint Laplace's method.

End of lecture 9